

# Sayantan Bhattacharya

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## EDUCATION

### Techno Main Salt Lake

B.Tech in Computer Science and Engineering (AI and Machine Learning)

Kolkata, India

CGPA: 7.23 / 10

Expected June 2027

- Relevant coursework: Machine Learning, Deep Learning, NLP, Neural Networks, Data Structures and Algorithms, Probability and Statistics, Linear Algebra, Database Management Systems, Web Development, Computer Networks, Operating Systems

## EXPERIENCE

### Independent Technical Contractor

2020 — 2025

Content Systems and Automation — Remote

- Automated a multi-stage post-production pipeline using scripted DaVinci Resolve workflows, reducing per-project processing time by 40% across 50+ client deliverables — applied systematic optimization and iterative testing to identify and eliminate bottlenecks in a complex, multi-input system.
- Managed end-to-end delivery of 30+ long-form projects across international clients by building a structured scoping and revision framework, achieving zero late deliveries over 5 years.
- Designed reusable modular templates for color grading, motion, and audio processing by abstracting recurring patterns into parameterized workflows, scaling output 3x without proportional time increase.

## PROJECTS

### PassME — Full-Stack Password Manager

2025

Python, Flask, SQLite, HTML, CSS, JavaScript

CS50x Final Project

- Achieved zero critical vulnerabilities across all OWASP Top 10 attack vectors by implementing bcrypt hashing, Fernet encryption, and parameterized queries throughout the authentication and storage pipeline.
- Reduced credential retrieval time to under 100ms by designing an indexed SQLite schema with optimized CRUD operations, supporting password generation, copy-to-clipboard, and session management.
- Delivered a fully functional, security-compliant password manager as a solo developer, verified through manual penetration testing of all input surfaces.

Demo: [youtube.com/watch?v=X2SUDguMxqc](https://youtube.com/watch?v=X2SUDguMxqc)

### Semantic Search RAG Chatbot — Two-Stage Retrieval Experiment

2026

Python, Flask, ChromaDB, LangChain, Groq (LLaMA-3.3 70B), Scrapy, HuggingFace Transformers

Personal Project

- Improved retrieval precision over single-stage cosine similarity search by designing a two-stage pipeline — ChromaDB vector search over 50 candidates followed by CrossEncoder reranking (ms-marco-MiniLM-L-6-v2) — with a Scrapy and Playwright crawler using SHA-256 hashing to avoid redundant re-indexing.
- Enabled coherent multi-turn interaction via a conversation memory module preserving the last 3 Q&A pairs, generating cited answers with LLaMA-3.3 70B via Groq API across PDF, URL, and crawled web sources.

GitHub: [github.com/PopcornKitsilano/rag-chatbot](https://github.com/PopcornKitsilano/rag-chatbot)

### BasketBawl — Basketball Hooping Game

2024

MIT Scratch

CS50x PSET0

- Built a real-time score-tracking game with keyboard-driven player movement (A/D) and shooting (1, 2 points per successful shot).
- Implemented conditional game flow with name-based prompts and graceful exit handling for non-affirmative user responses.

Link: [scratch.mit.edu/projects/1061387522](https://scratch.mit.edu/projects/1061387522)

Razer Crimson Royale — Custom Hardware Lighting Engine

Python, Razer Chroma REST API

2026

Personal Project

- Built a real-time per-key animation engine delivering 20 effects at 7fps by implementing HSV color math, directional transforms, and a session-based REST protocol — demonstrating applied systems programming outside coursework.
- Eliminated Razer Synapse dependency by reverse-engineering the local Chroma SDK REST API, implementing BGR encoding and a terminal-driven playlist system with full runtime control over effect order, speed, and direction.

GitHub: [github.com/PopcornKitsilano/razer-crimson-royale](https://github.com/PopcornKitsilano/razer-crimson-royale)

TECHNICAL SKILLS

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**Languages:** Python, JavaScript, C, C++, Java, SQL, Bash

**AI and ML:** Machine Learning, Deep Learning, NLP, Computer Vision, Neural Networks, LLM frameworks, LangChain, Prompt Engineering, Model development and deployment, AWS SageMaker, AWS Bedrock, Conversational AI, Generative AI, Supervised and Unsupervised Learning, Feature Engineering

**ML Libraries:** NumPy, pandas, scikit-learn, Matplotlib, TensorFlow, PyTorch, Hugging Face Transformers

**Web and Frameworks:** Flask, Django, React, Node.js, HTML5, CSS3, Bootstrap, REST APIs

**Databases:** SQLite, SQL (schema design, indexing, joins, CRUD operations)

**Cybersecurity:** Penetration testing, OWASP Top 10, vulnerability assessment, SIEM, log analysis

**Security Tools:** BurpSuite, Metasploit, Wireshark, Hydra, CAPA, Splunk, Graylog

**Developer Tools:** Git, GitHub, Linux (Arch, Kali), DaVinci Resolve, Arduino, Agile, VS Code

**Concepts:** OOP, Data Structures and Algorithms, Software Design, Secure Coding

CERTIFICATIONS

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| <b>CS50X — Introduction to Computer Science</b><br>Harvard University via edX   <a href="#">View Certificate</a>   <a href="#">Verified PDF</a> | Aug 2025 |
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| <b>Getting Started with Artificial Intelligence</b><br>IBM via Credly   <a href="#">View Certificate</a> | Nov 2024 |
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| <b>PreSecurity Certificate</b><br>TryHackMe — 100+ rooms; real-world attack simulation, privilege escalation   <a href="#">View Certificate</a> | Jul 2025 |
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TEST SCORES

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- SAT (Scholastic Assessment Test): 1540 / 1600 | College Board | Dec 2021

HOBBIES

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Basketball | Music Sampling | [Making YouTube Videos](#)